SHEAFIFICATION COMMUTES WITH FINITE LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Exactness of left adjoints. Categories, Lemma categories-lemma-exact-adjoint. Gives right exactness of sheafification.
- The colimits in the plus construction are directed. Same chapter, Lemmas lemma-common-refinement and lemma-independent-refinement. Identify the indexing category of the colimits in the plus construction with a directed set.
- Filtered colimits commute with finite limits. Categories, Lemma categories-lemma-directed-commutes. Supplies the commutation with finite limits for the plus construction.
- Limits of sheaves are computed as presheaves. Same chapter, Lemma lemma-limit-sheaf. Transfers the statement from presheaves to sheaves.

Source reference. Original lemma 0039: sources/stacks-sites/pinned-original-source.tex, source lines 2358–2358. Navigation pointer invoked at proof line 2358; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00W6: sources/stacks-sites/pinned-original-source.tex, source lines 2359–2359. Navigation pointer invoked at proof line 2359; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00W7: sources/stacks-sites/pinned-original-source.tex, source lines 2360–2360. Navigation pointer invoked at proof line 2360; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 002W: sources/stacks-sites/pinned-original-source.tex, source lines 2365–2365. Navigation pointer invoked at proof line 2365; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00W2: sources/stacks-sites/pinned-original-source.tex, source lines 2369–2369. Navigation pointer invoked at proof line 2369; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *The functor $PSh(\mathcal{C}) \to Sh(\mathcal{C})$, $\mathcal{F} \mapsto \mathcal{F}^\#$ is exact.*

Proof. Since it is a left adjoint it is right exact, see Categories, Lemma 0039. On the other hand, by Lemmas 00W6 and Lemma 00W7 the colimits in the construction of $\mathcal{F}^+$ are really over the directed set $\mathrm{Ob}(\mathcal{J}_U)$ where $\mathcal{U} \geq \mathcal{U}'$ if and only if $\mathcal{U}$ is a refinement of $\mathcal{U}'$. Hence by Categories, Lemma 002W we see that $\mathcal{F} \to \mathcal{F}^+$ commutes with finite limits (as a functor from presheaves to presheaves). Then we conclude using Lemma 00W2. $\square$